

WEEK - 2

DATA MODELING, DAX & SALES DASHBOARD

Introduction :

This project focuses on creating an interactive sales analysis dashboard using Power BI. The SampleSuperstore dataset was used to analyze sales, profit, orders and quantity. The project includes data modeling, DAX measures, visualizations and interactive filtering. The final dashboard presents the data in a simple, clear and meaningful way.

Objectives :

The main objectives of this project are:

- To import and understand the SampleSuperstore dataset.
- To create supporting tables for better data organization.
- To establish relationships between the tables.
- To create DAX measures for important calculations.
- To design an interactive Power BI dashboard.
- To represent data using KPI cards and charts.
- To implement slicers and cross-filtering.
- To identify useful insights from the sales data.

Dataset Description :

The SampleSuperstore dataset contains information related to customers, products, orders and sales transactions. Important columns include Customer ID, Customer Name, Segment, Product ID, Product Name, Category, Sub-Category, Sales, Profit, Quantity, Order ID and Order Date.

This dataset was used to analyze overall business performance and category-wise sales performance.

Data Modeling :

Data modeling was performed to organize the dataset and establish relationships between the tables.

A DimCustomer table was created using DAX. It contains Customer ID, Customer Name and Segment information. A DimProduct table was created using Power Query by referencing the SampleSuperstore table and keeping the required product-related columns.

The following relationships were created:

- DimCustomer → SampleSuperstore using Customer ID
- DimProduct → SampleSuperstore using Product ID

Both relationships are active one-to-many relationships. This creates an organized data model for analysis.

DimCustomer :

Customer ID	Customer Name	Segment
LS-17230	Lyoris Saunders	Consumer
MV-17485	Mark Van Huff	Consumer
JL-15835	John Lee	Consumer
FP-14320	Frank Preis	Consumer
KD-16345	Katherine Ducich	Consumer
CC-12145	Charles Crestani	Consumer
CB-12025	Cassandra Brandow	Consumer
JG-15160	James Galang	Consumer
CL-12565	Clay Ludtke	Consumer
EH-13945	Eric Hoffmann	Consumer
JO-15280	Jas O'Carroll	Consumer
AS-10240	Alan Shonely	Consumer
GT-14755	Guy Thornton	Consumer
XP-21865	Xylona Preis	Consumer
TB-21250	Tim Brodman	Consumer
GM-14440	Gary McGarr	Consumer
DD-13570	Dorothy Dickinson	Consumer
RL-19615	Rob Lucas	Consumer
WB-21850	William Brown	Consumer

Table: DimCustomer (804 rows)

DimProduct :

Product ID	Category	Sub-Category	Product Name
OFF-PA-10000174	Office Supplies	Paper	Message Book, Wirebound, Four 5 1/2" X 4" Forms/Pg., 200 Dupl. Sets/Book
OFF-PA-10000955	Office Supplies	Paper	Southworth 25% Cotton Granite Paper & Envelopes
OFF-PA-10002005	Office Supplies	Paper	Xerox 225
OFF-PA-10002947	Office Supplies	Paper	Xerox 1923
OFF-PA-10000659	Office Supplies	Paper	TOPS Carbonless Receipt Book, Four 2-3/4 x 7-1/4 Money Receipts per Page
OFF-PA-10002893	Office Supplies	Paper	Wirebound Service Call Books, 5 1/2" x 4"
OFF-PA-10002689	Office Supplies	Paper	Weyerhaeuser First Choice Laser/Copy Paper (20Lb. and 88 Bright)
OFF-PA-10001804	Office Supplies	Paper	Xerox 195
OFF-PA-1000380	Office Supplies	Paper	REDIFORM Incoming/Outgoing Call Register, 11" X 8 1/2", 100 Messages
OFF-PA-10003072	Office Supplies	Paper	Eureka Recycled Copy Paper 8 1/2" x 11", Ream
OFF-PA-10001204	Office Supplies	Paper	Xerox 1972
OFF-PA-1000350	Office Supplies	Paper	Message Book, Standard Line "While You Were Out", 5 1/2" X 4", 200 Sets/Book
OFF-PA-1000466	Office Supplies	Paper	Memo Book, 100 Message Capacity, 5 3/8" x 11"
OFF-PA-10003363	Office Supplies	Paper	Xerox 204
OFF-PA-10001800	Office Supplies	Paper	Xerox 220
OFF-PA-10000069	Office Supplies	Paper	TOPS 4 x 6 Fluorescent Color Memo Sheets, 500 Sheets per Pack
OFF-PA-10003036	Office Supplies	Paper	Black Print Carbonless 8 1/2" x 8 1/4" Rapid Memo Book
OFF-PA-10000327	Office Supplies	Paper	Xerox 1971
OFF-PA-10000587	Office Supplies	Paper	Array Parchment Paper, Assorted Colors
OFF-PA-10001878	Office Supplies	Paper	Xerox 1891
OFF-PA-10002259	Office Supplies	Paper	Geographics Note Cards, Blank, White, 8 1/2" x 11"
OFF-PA-10000213	Office Supplies	Paper	Xerox 198
OFF-PA-10002137	Office Supplies	Paper	Southworth 100% Résumé Paper, 24lb.
OFF-PA-10000176	Office Supplies	Paper	Xerox 1887
OFF-PA-10001215	Office Supplies	Paper	Xerox 1963
OFF-PA-10000170	Office Supplies	Paper	Xerox 1960

DAX Measures Created :

The following DAX measures were created for the dashboard.

Total Sales :

Total Sales = SUM(SampleSuperstore[Sales])

Total Profit :

Total Profit = SUM(SampleSuperstore[Profit])

Total Orders :

Total Orders = DISTINCTCOUNT(SampleSuperstore[Order ID])

Total Quantity :

Total Quantity = SUM(SampleSuperstore[Quantity])

Average Sales :

Average Sales = AVERAGE(SampleSuperstore[Sales])

Profit Margin % :

Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)

These measures were used to display important business metrics in the dashboard.

Dashboard / Visualizations :

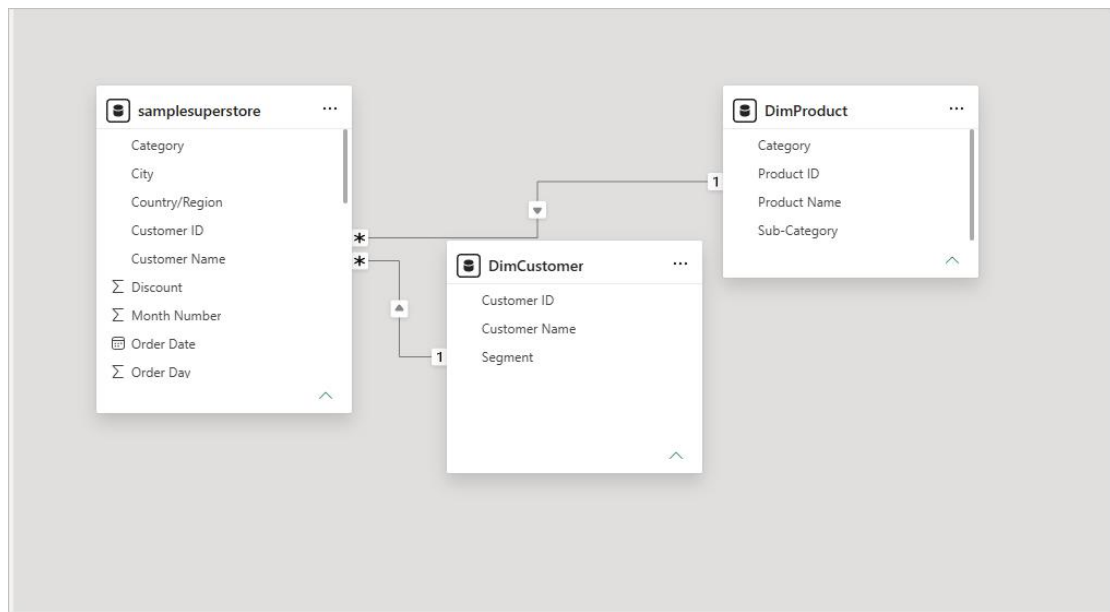
An interactive dashboard was created using different Power BI visuals. The dashboard contains six KPI cards:

- **Total Sales – 2.33M**
- **Total Profit – 292.30K**
- **Total Orders – 5K**
- **Total Quantity – 39K**
- **Average Sales – 228.23**
- **Profit Margin – 12.56%**

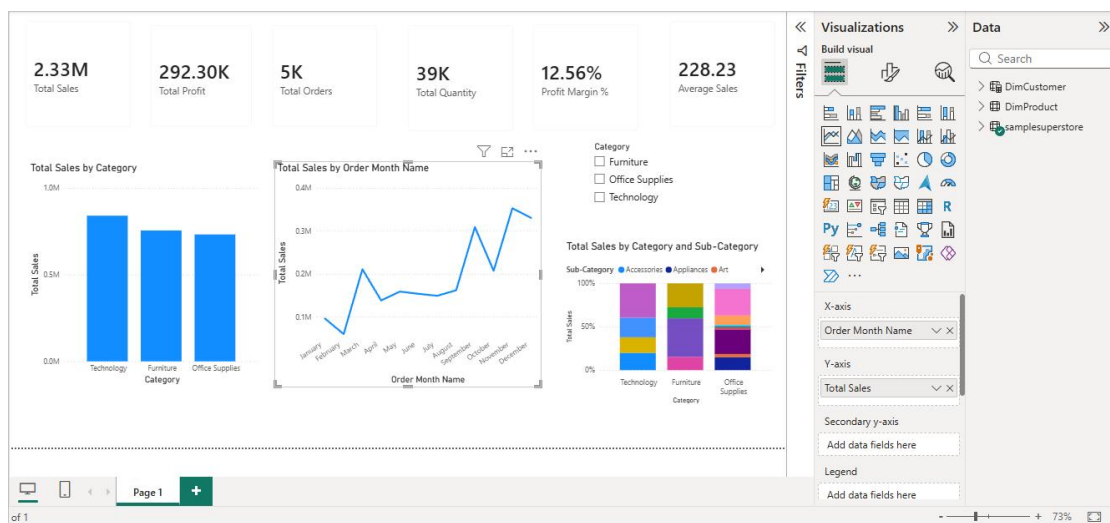
The dashboard also contains a Total Sales by Category chart, a Total Sales by Order Month line chart and a Stacked Column Chart showing sales based on Category and Sub-Category.

The monthly sales chart was arranged in the correct January to December order using the Month Number column for sorting.

Screenshot 2 – Data Model and Relationships



Screenshot 3 – Final Dashboard Output :



Slicer & Cross-Filtering :

A Category Slicer was added to make the dashboard interactive. The slicer contains three categories:

- Office Supplies
- Furniture
- Technology

When a category is selected, the KPI cards and charts automatically update according to the selected category. This shows that the slicer and cross-filtering functionality are working correctly.

The interactive filtering helps users easily compare the performance of different categories.

Key Insights :

The dashboard provides the following key insights:

- The overall Total Sales is approximately 2.33M.
- The overall Total Profit is approximately 292.30K.
- The Total Orders are approximately 5K.
- The Total Quantity is approximately 39K.
- The Average Sales is approximately 228.23.
- The Profit Margin is approximately 12.56%.
- Sales performance can be compared across different categories.
- Monthly sales can be analyzed from January to December.
- The Category Slicer allows users to view category-specific performance.
- KPI cards and charts provide a quick overview of the business performance.

Conclusion :

The Power BI Week 2 project was successfully completed by creating a structured data model, establishing relationships, developing DAX measures and designing an interactive dashboard. The use of KPI cards, charts, slicers and cross-filtering makes the dashboard simple, clear and user-friendly.

The completed dashboard provides an effective way to analyze sales, profit, orders, quantity and category performance using Power BI.